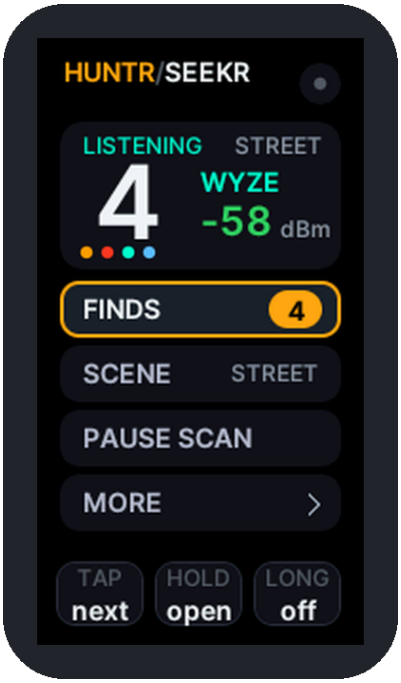
HUNTR/SEEKR

Counter-surveillance signal scanner

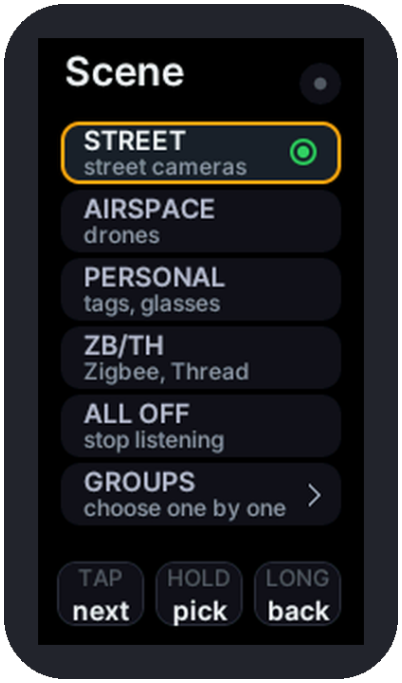
One-button Edition
User Guide and Start-up Guide

Waveshare ESP32-C6-LCD-1.47 (plain, no touch)

Public Beta 1



Home



Pick a scene



Track a signal

Printable guide. Screen pictures are example renderings of the interface produced by a simulator, not photographs of the board.

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1. About this device

HUNTR/SEEKR turns the **Waveshare ESP32-C6-LCD-1.47** into a pocket radio scanner. It listens on the 2.4 GHz band for the wireless signals that certain kinds of devices give off, compares them with built-in lists of known makers and names, and shows you what it heard and how strong each signal is. You can then follow a signal toward its source.

This edition is for the **plain** board, the one with **no touch panel**. The only button the software can read is **BOOT**, so the whole interface is built around that one button (section 4). If your board has a touch screen (its name has **Touch** in it), use the touch edition of the software instead.

It is **passive**: it only receives. It does not jam, attack, or connect to anyone's network. The one thing it transmits is its own small Wi-Fi hotspot, and only while you have switched the phone page on (section 6.8).

> **IMPORTANT Results are hints, not proof**
> Names and addresses can be changed or randomized, many devices are not in the lists, and a quiet device may simply be out of range. "Nothing found" never proves nothing is there. A match is a clue about a device, never evidence about a person or a place.

1.1 What it can find

Kind	Examples	How it is spotted
Wi-Fi devices	Security cameras, doorbells, smart glasses, drones, licence-plate readers	Network names and hardware addresses
Rogue Wi-Fi signs	De-authentication floods, open "evil twin" copies of a network name, one transmitter answering to many names	Flagged as warnings, not proof
Bluetooth LE devices	Trackers (AirTag and other Find My trackers, Tile, SmartTag, Google Find Hub, Chipolo, Pebblebee), body cameras, smart glasses, drones, beacons, wearables, smart lights, Fast Pair accessories	Names, makers and service codes; each gets a type tag
Zigbee and Thread	Smart-home radios (802.15.4)	The chip's own 802.15.4 radio
Drone Remote ID	The serial number and position many drones broadcast	Wi-Fi and Bluetooth Remote ID messages
Possible card skimmers	A Bluetooth module still wearing a factory name, the kind sometimes hidden in gas pumps and ATM readers	A clue, not a verdict

1.2 What this edition does not have

The plain board has no touch panel, no motion sensor, no battery gauge, no clock chip and no speaker. So compared with the touch edition this one has **no direction finder (AIM)**, no landscape screen, no session-report or self-test screens, and no sound unless you wire a buzzer (section 10). The detection side, which decides what is found and how it is shown, is the same code. The things that need a finger (settings pages, ignore-list editing, calibration) are not in this edition; section 11 lists them.

2. What you need

Item	Why
The Waveshare ESP32-C6-LCD-1.47 board	The board this guide covers: a 1.47 inch screen (172 x 320), an RGB LED, a BOOT button, a RESET button, a USB-C port and a TF card slot (not used by this software). 4 MB of flash.
A USB data cable	Some cables only charge. If the computer does not see the board, try another cable.
A computer (Windows, macOS or Linux)	To put the software on the board.
A phone (optional)	Used for the live view, to set the clock (the board has no clock chip), to add your own signatures, to download saved logs and to choose a boot picture.
A passive buzzer (optional)	For sound. There is no speaker on the board (section 10).
Up to two extra push buttons (optional)	For a quicker interface (section 10).
A GPS module (optional)	Connects to the UART pads and adds your position to the log.

> **NOTE Power**
> The board runs from USB power. It has no battery gauge, so the software shows no battery level. If you power it some other way, follow the board maker's instructions.

3. Putting the software on the board

There are two ways: write the ready-made **.bin** file (the easy way), or build it yourself from the source.

3.1 Write the .bin file

1. Download **HUNTR-SEEKR_C6_LCD147_BUTTON_Public_Beta_1.bin** and **SHA256SUMS.txt** from the project's Releases page.
2. Install Python 3, then run **pip install esptool** (on Windows, if pip is not found, use **py -m pip install esptool**).
3. Plug the board into the computer with the USB-C cable and find its port: on Windows a COM port in Device Manager (for example COM5), on macOS **/dev/cu.usbmodem...**, on Linux **/dev/ttyACM0**.
4. Run: **esptool --chip esp32c6 --port YOUR_PORT --baud 460800 write-flash 0x0 HUNTR-SEEKR_C6_LCD147_BUTTON_Public_Beta_1.bin**
5. When it says *Hash of data verified* and *Hard resetting* the board restarts by itself. The boot screen ends with **ready Public Beta 1**.

Writing the file replaces the software and resets the saved settings. Logs saved on the board are kept. To check the download first, compare its SHA-256 with the line in SHA256SUMS.txt (Windows PowerShell: **Get-FileHash .\file.bin -Algorithm SHA256**; macOS and Linux: **shasum -a 256 -c SHA256SUMS.txt**).

> **TIP If the computer cannot connect**
> Use a data cable and try another USB port. Then put the chip in download mode: **unplug the cable, hold BOOT, plug the cable in, release BOOT**, and run the command again. If the connection is unreliable add **--baud 115200**.

The project's web flasher (**skirf-labs.tech**) writes the same file from Google Chrome or Microsoft Edge on a computer, with nothing to install.

3.2 Build it from the source

Open **src/HUNTR-SEEKR_C6_LCD147_BUTTON/HUNTR-SEEKR_C6_LCD147_BUTTON.ino** in the Arduino IDE (the three files in that folder must stay together). Install the **esp32** board package by Espressif (3.3.5 or newer) and the libraries **GFX Library for Arduino** and **TinyGPSPlus**. Then choose:

Setting	Value
Board	ESP32C6 Dev Module
USB CDC On Boot	Enabled (so the Serial Monitor works at 115200 baud)
Flash Size	4MB (32Mb)
Partition Scheme	Huge APP (3MB No OTA/1MB SPIFFS). This layout has no update slot: the board is updated over USB.

Press Upload. docs/BUILDING.md in the project has the command-line build and the steps that turn a build into the release file.

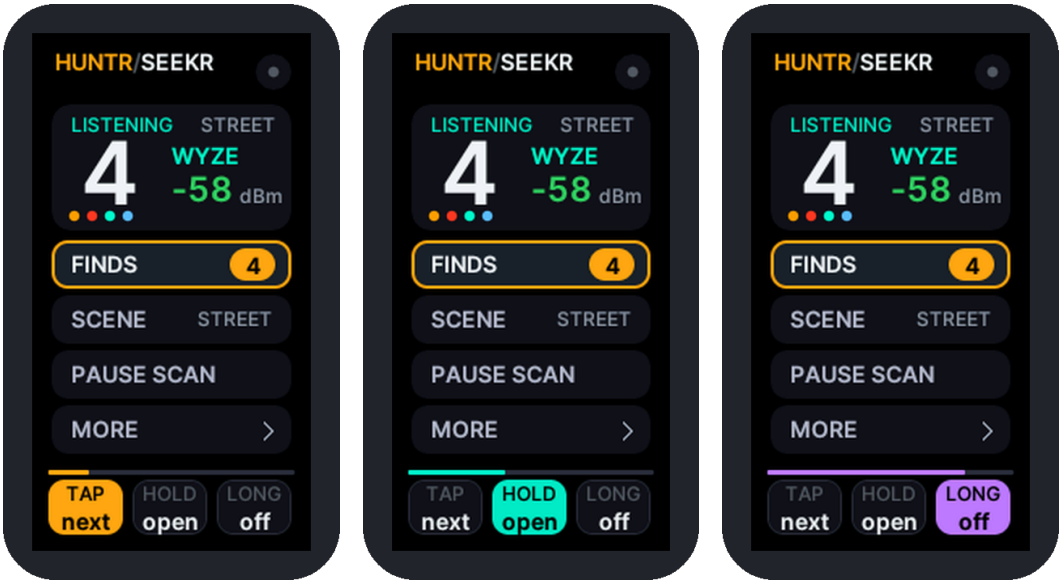
3.3 First start-up checklist

- The screen lights and shows the boot picture: a radar and the word HUNTR/SEEKR, then **ready Public Beta 1** at the bottom. A short press of the button skips it.
- The Home screen appears (section 6.1). The RGB LED is dark until you arm something.
- Press the button once and the highlight moves down a row: the button works.
- If the screen stays black or the picture looks wrong, see section 12.

4. Using the one button

Everything is done with the **BOOT** button, the small button on the board marked BOOT (it is wired to pin IO9). The **RESET** button only restarts the chip; the software cannot read it. The button reads **how long you hold it**, and the action happens **when you let go**.

Gesture	Button held for	What it does
TAP	under 0.5 seconds	**NEXT**: move to the next item in the list
HOLD	0.5 to 1.5 seconds	**SELECT**: open, start or apply the highlighted item
LONG	over 1.5 seconds	**BACK**: go back one step. On the Home screen: turn the screen off



Under 0.5 s. The TAP box lights. **0.5 to 1.5 s.** The HOLD box lights. **Over 1.5 s.** The LONG box lights.

4.1 The hint strip

The three boxes along the bottom of every screen say **what TAP, HOLD and LONG will do right now**. While the button is down, a bar sweeps across above them and the box for the gesture you are making lights up, so you can see which one you are about to make: let go while the box you want is lit. A dash means that gesture does nothing on this screen.

4.2 The LED helps too

The RGB LED changes colour at the same moments: **teal-green** once you have held past half a second (SELECT), **violet** once you have held past one and a half seconds (BACK). So the button also works by feel and by light with the screen in a pocket.

4.3 Good to know

- **Wake-up press.** When the screen is off or dimmed, the first press only wakes it and does nothing else. The next press acts.
- **Switching on.** A button that is already held when the board powers up does nothing until it has been let go once.
- **Stuck button.** A press that lasts longer than 9 seconds is treated as a stuck button and ignored until it is released.
- **Contact bounce** is filtered out: a single press is always one gesture.
- **Cut-offs.** The 0.5 s and 1.5 s limits are **BTN_SEL_MS** and **BTN_BACK_MS** near the top of the sketch, if you want them quicker or slower.

5. First use, step by step

1. At power-up the **Home** screen appears and nothing is armed (it says IDLE). The highlight is on **FINDS**.
2. **TAP** twice to move the highlight to **SCENE**, then **HOLD** to open the scene list.
3. **TAP** to the scene you want (for example STREET, for street cameras) and **HOLD** to arm it. Listening starts at once and Home says **LISTENING**. The LED gives a slow teal blip every few seconds.
4. When something is found, a **banner** drops down from the top and the LED pulses in the group's colour. **TAP** skips the banner, **HOLD** tracks that device (section 6.5).
5. To look at everything that has been heard, go to **FINDS** and **HOLD**. **TAP** walks down the list, **HOLD** tracks the highlighted device, **LONG** goes back.
6. For the pocket: **LONG** on Home turns the screen off. Scanning and the LED carry on, and any press wakes the screen. The scene you chose comes back by itself after a restart.

> **TIP A good habit**
> Pick the scene that matches what you care about, rather than arming everything. Fewer groups means each one is heard more reliably, and the Wi-Fi and Bluetooth radios share one antenna.

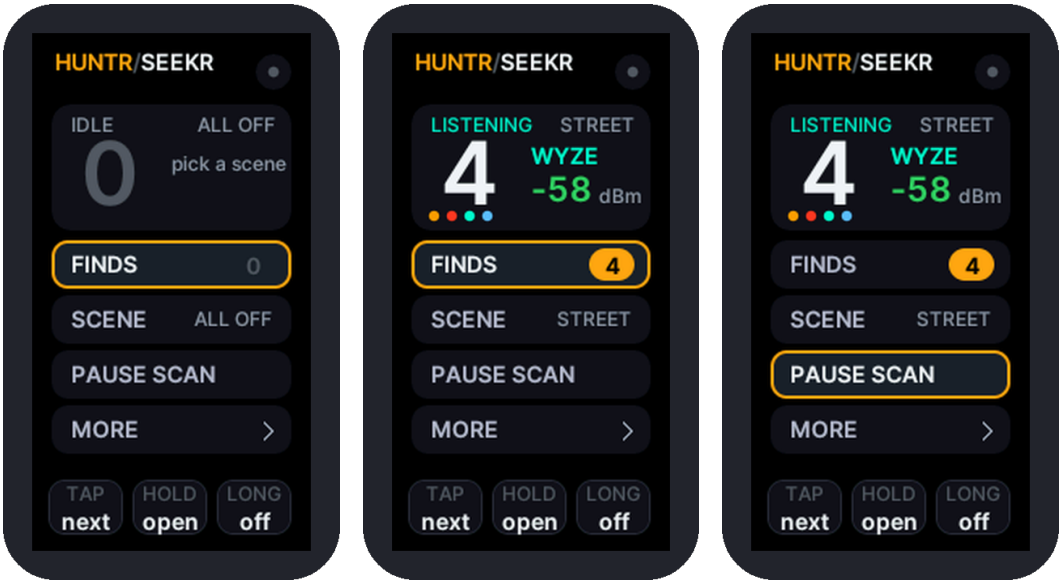
6. The screens

6.1 Home

The top card shows the state (**IDLE**, **LISTENING** or **PAUSED**), the current scene, a big count of finds, the strongest find with its signal in dBm, and one coloured dot for each armed group. Four rows sit below it:

Row	TAP / HOLD / LONG
FINDS (with a count)	HOLD opens the list of everything heard (section 6.4).
SCENE (with the scene name)	HOLD opens the scene list (section 6.2).
PAUSE SCAN / RESUME SCAN	HOLD pauses or resumes listening. With nothing armed it just says to pick a scene first.
MORE	HOLD opens the More menu (section 6.7).

On Home, **TAP** moves down a row (after the last row it goes back to the first), and **LONG** turns the screen off.



Home, nothing armed.

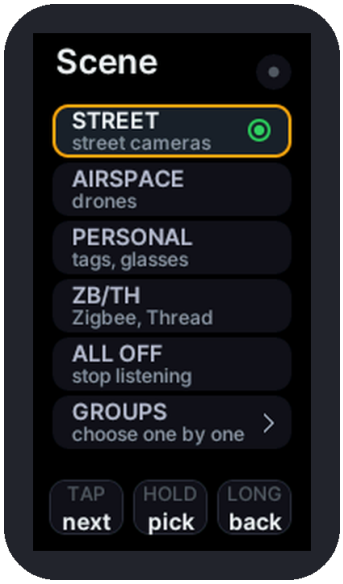
Home, listening. The count, the strongest find, the group dots.

PAUSE SCAN highlighted.

6.2 Scene list

A **scene** arms several groups at once. HOLD on a scene arms it (the green mark shows the active one) and returns to Home; choosing another scene replaces the first. The last row, **GROUPS**, lets you pick groups one by one (section 6.3).

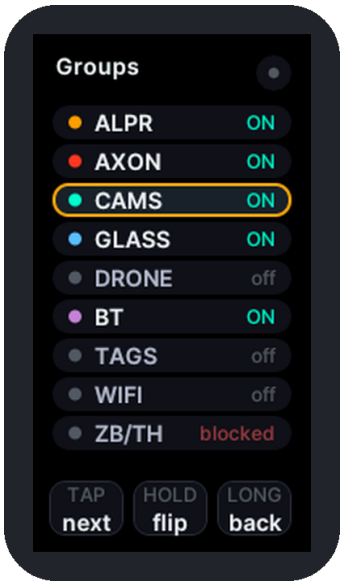
Scene	Arms
STREET	ALPR, AXON, CAMS, GLASS: street cameras
AIRSPACE	DRONE: drones
PERSONAL	BT, TAGS: trackers, glasses, wearables
ZB/TH	Zigbee and Thread (the 802.15.4 radio)
ALL OFF	Nothing: stop listening



Scene list. TAP next, HOLD pick, LONG back.

6.3 Groups

The group list shows every detection group with **ON** or **off**. **TAP** walks down it, **HOLD** flips the highlighted group, **LONG** goes back to the scene list. A set you pick by hand shows as **CUSTOM** on Home. Section 7 says what each group listens for.



Groups. Flip them one at a time. A red **blocked** means the other radio side is in the way (section 7).

6.4 Finds

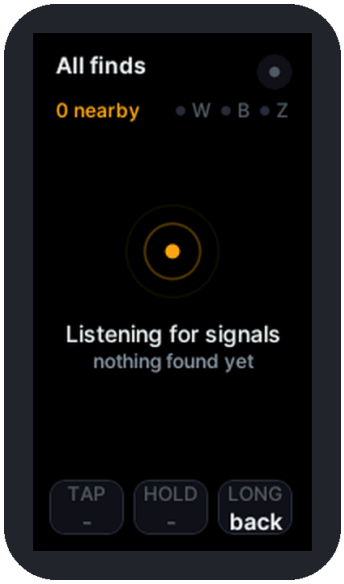
The list of everything heard, strongest first, with the group colour, the name, the signal bars and the radio (WIFI or BLE). The number under the title says how many are nearby. **TAP** moves the highlight down and wraps to the top at the end, **HOLD** starts tracking the highlighted device, **LONG** goes back. The list scrolls by itself to keep the highlighted row on screen. An empty list says *Listening for signals*.



Finds. The highlighted row has the amber frame.



Scrolled. The highlight stays on screen.

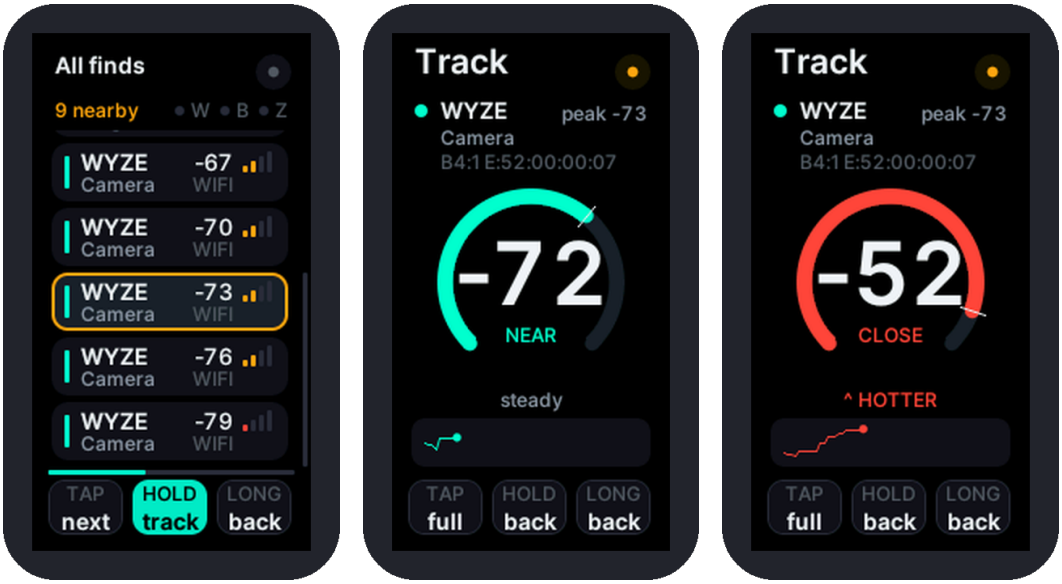


Nothing yet.

6.5 Track

Track follows one device. A big signal number (in dBm) sits inside a ring: the closer the number is to zero, the closer the device. The ring fills as you get nearer, and it is zoomed to the range that matters, so small changes show. The word under the number is **FAR**, **NEAR** or **CLOSE**. Under the ring, **HOTTER** or **COLDER** tells you whether the last few readings are getting stronger or weaker, and the small graph at the bottom is the recent history. **peak** is the strongest reading so far.

TAP switches between the zoomed ring and the full range (the box says what it will switch to). **HOLD** or **LONG** leaves Track. If you have fitted a buzzer (section 10), TAP turns the sound on and off instead, and beeps get faster as you get closer. The LED blinks faster as you get closer too (section 8), so you can track with the screen away from your eyes.



Holding on a find. The strip fills: let go to track.

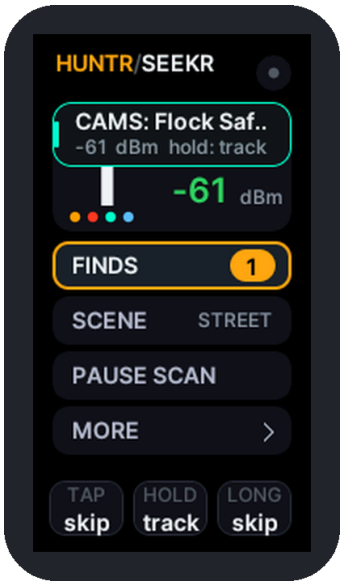
Track. Zoomed ring and number.

Close. The ring turns red.

> **NOTE Signal strength is a rough guide**
> Walls, bodies and reflections change a reading from one step to the next. Use the number to compare as you move: warmer means closer. It cannot give a distance or a direction by itself. This edition has no direction finder; the touch edition has one.

6.6 The find banner

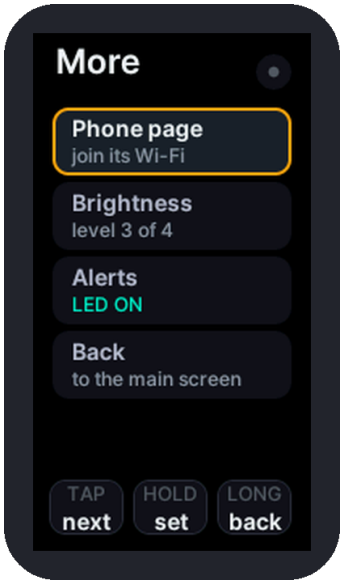
When something new is found, a banner slides in below the title with the group and the name, and the LED pulses in the group's colour. **TAP** or **LONG** skips it, **HOLD** tracks that device (the banner says *hold: track*). A tracker that has been following you, a possible card skimmer and the rogue Wi-Fi warnings use the same banner, in red. While you are tracking a device a new banner never takes over: HOLD still just leaves.



A find. TAP skips it, HOLD tracks it.

6.7 More

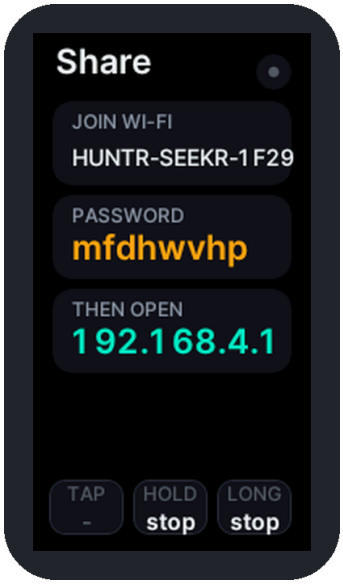
Row	What HOLD does
Phone page	Switches on the board's Wi-Fi hotspot and shows the Share screen (section 6.8).
Brightness	Steps the screen through four levels. The choice is remembered.
Alerts	Cycles **LED ON**, **LED OFF** and **DISCREET** (with a buzzer fitted: LED + SOUND, LED ONLY, SOUND ONLY, NO ALERTS, DISCREET). Discreet mode switches off the LED, the sound and the banners, and dims the screen after 30 seconds and turns it off after two minutes. The choice is remembered.
Back	Returns to Home.



More. TAP next, HOLD set, LONG back.

6.8 The phone page (Share)

The phone page lets a phone talk to the board over the board's own Wi-Fi. HOLD on **More > Phone page** starts a hotspot and shows its **name** (**HUNTR-SEEKR-** followed by four characters that change every time), a random **password** and the address **192.168.4.1**. Join that network with your phone, then open **http://192.168.4.1** (not https) in the phone's browser. While the hotspot is on, Wi-Fi scanning pauses and the LED blinks blue slowly. **HOLD** or **LONG** stops it and returns to More; TAP does nothing there.



Share. Join the Wi-Fi shown, then open the address.

On the phone page	What it does
Live view	A live list of what the board is hearing. Opening it also sets the board's clock from the phone, so saved logs get time stamps.
Signatures	Add your own detection rules (up to 128) and a signature update file (sigs_update.txt).
Boot logo	Upload your own picture for the boot screen. When it is saved the boot screen uses it from then on.
Files	The saved logs on the board, each one a link to download. There is also a button that wipes all logs.
Update firmware	Not available in this edition: its layout has no update slot. Update over USB (section 3).

7. The detection groups

Group	What it listens for
ALPR	Licence-plate readers
AXON	Axon body cameras and related gear
CAMS	Consumer cameras: doorbells, Wyze, Ring and similar, and other security cameras
GLASS	Smart glasses with cameras
DRONE	Drones, by their Wi-Fi names and by Remote ID messages
BT	Generic Bluetooth devices: beacons, wearables, smart lights, Fast Pair accessories
TAGS	Tracker tags: AirTag and other Find My trackers, Tile, SmartTag, Google Find Hub, Chipolo, Pebblebee
WIFI	Other Wi-Fi devices of interest, and the rogue Wi-Fi warnings
ZB/TH	Zigbee and Thread smart-home radios, using the chip's 802.15.4 radio

7.1 "Blocked"

The chip has one radio. It cannot do Wi-Fi scanning and 802.15.4 (Zigbee and Thread) at the same time, so **ZB/TH is refused while any Wi-Fi group is on**, and Wi-Fi groups are refused while ZB/TH is on. The screen tells you which side is in the way: turn that side off first. Wi-Fi and Bluetooth groups can run together; each hears a little less than it would alone, because they share the radio.

7.2 What a find shows

- A coloured stripe and dot for the group, the device's name or maker, the signal in dBm and bars, and the radio (WIFI or BLE).
- **Type tags** for Bluetooth finds: IBEACON, EDDYSTONE, ALTBEACON, FASTPAIR, MESH, LIGHT, WEARABLE and SKIMMER.
- **FOLLOW** on a tracker that has stayed in range with you for a long time (the tracker "following" alert).
- A red **SKIMMER?** banner for a Bluetooth module still wearing a factory name of the kind used in card skimmers. This is a clue, not a verdict: many harmless things use the same modules.

7.3 Other detection features

These work in the background in the same way as on the touch edition: Bluetooth spam detection, rogue Wi-Fi checks (DEAUTH, TWIN and KARMA), fixed-address probe listing, device fingerprint codes, drone Remote ID decoding, and your own signature rules. The settings and review screens for some of them need a touch screen and are not in this edition (section 11).

8. The RGB LED

The LED on the board is the status light. It is capped to a modest brightness because it is very bright through the clear cover (**LED_LEVEL** in the sketch). Turn it off in **More > Alerts**.

What you see	Meaning
Slow teal blip every 3 seconds	Listening, nothing new
Slow amber blip every 6 seconds	Armed but paused
Pulse in a group colour (ALPR amber, AXON red, CAMS teal, GLASS blue, DRONE yellow, BT purple)	A find in that group: four blinks a second at first, then a calm breathing until the banner goes
Keeps breathing in that colour	A find arrived while the screen was off, and no button has been pressed since
Blinks faster as you get closer; solid when you are on top of it	While tracking. Blue is far, amber is near, red is close
Slow blue blink	The phone page (Share) is on
Teal-green, then violet, while the button is held	The HOLD and LONG boundaries (section 4.2)
Dark	Discreet mode, or the LED is switched off in More > Alerts

9. Logs, signatures and the clock

9.1 Saved findings

This edition saves what it has found **automatically**: the findings record is written to the board's flash every five minutes while there is something new, and again when scanning stops (you pause it, pick ALL OFF, or switch the board off after a pause). Each session makes a new file. A record shows each device once, with its group, name, signal and, if a GPS module is fitted and has a position, where it was.

To get the files, use the phone page (section 6.8): open **Files**, tap a file to download it. The same page can wipe all logs when the flash is getting full (about 0.9 MB is available for storage; the board says *Storage full: log not saved* when it runs out).

9.2 The clock

The board has no clock chip. Open the phone page's **Live view** once after each start: the board takes the time and time zone from the phone, so log files get time stamps. Until then times in logs are blank.

9.3 Your own signatures

On the phone page, **Signatures** lets you add your own rules: oui (hardware-address prefix), ssid (network name), ble (Bluetooth name), cid (company code), uuid (service code) and ie rules, up to 128. A **sigs_update.txt** file uploaded from the same page is read after them. A match shows in the list with the tag **CUSTOM**.

9.4 A boot picture of your own

Boot logo on the phone page takes a picture and uses it for the boot screen from then on. Follow the instructions on that page for the size and format. Use **Delete** there to go back to the built-in radar picture.

10. Optional extras

10.1 Extra buttons

You can wire up to **two normally-open push buttons**, each between a free header pin and **GND** (no resistor needed). Put the pin numbers in **PIN_BTN_EXT1** and **PIN_BTN_EXT2** near the top of the sketch and upload again. Then **EXT1 = NEXT**, **EXT2 = SELECT**, and **BOOT = BACK**, and each acts the instant it is pressed, with no timing. The hint strip then labels the boxes BTN1, BTN2 and BOOT. Free pins on this board: **IO0, IO1, IO2, IO3, IO18, IO19, IO20, IO23**. Avoid IO12 and IO13 (USB) and IO4 (the TF card select line).

10.2 A buzzer

The board has no speaker. Wire a **passive** buzzer or piezo between a free header pin (for example IO3) and GND and put that pin number in **PIN_BUZZER** (it is -1, meaning none, by default). Then the alert tunes play, Track beeps get faster as you get closer, and More > Alerts gains sound choices. In Track, TAP then turns the sound on and off.

> **IMPORTANT Pins to keep away from**
> IO6 and IO7 carry the display's data and clock, IO14 and IO15 the display's select and data/command lines, IO21 and IO22 its reset and backlight, IO8 the RGB LED and IO9 the BOOT button. Do not put a buzzer or button on those.

10.3 A GPS module

A GPS module (9600 baud) connects to the UART pads: the module's TX to the board's **IO17** (RX) and its RX to **IO16** (TX). With a fix, logs carry positions.

10.4 Settings in the sketch

Name	Default	What it does
LCD_IPS	1	Colours inverted the way an IPS panel needs. If the picture looks like a photo negative, set 0.
LCD_COL_OFS	34	A 172 px wide panel sits 34 columns into the controller's 240 px memory. If the picture is shifted sideways or has a coloured edge, try 0 or 35.
LED_LEVEL	64	RGB LED brightness cap, 0 to 255.
BTN_SEL_MS, BTN_BACK_MS	500, 1500	The TAP / HOLD / LONG cut-offs in milliseconds.
PIN_BTN_EXT1, PIN_BTN_EXT2	-1, -1	Optional extra buttons (section 10.1).
PIN_BUZZER	-1	Optional buzzer pin (section 10.2).

11. What this edition does not do

These features of the touch edition need a touch screen, a keyboard or a motion sensor, so they are not in this edition:

- The direction finder **AIM** and its calibration (it needs the motion sensor).
- Landscape and rotated screens, the Auto-rotate and motion features (raise to wake, shake to pause).
- The **SETUP** pages: ignore list editing, the session report, the self-test, the Look settings, Bluetooth share, and the other individual settings. The defaults are used.
- The SWEEP page (walkthrough rooms and baselines) and the LOG page's SAVE LOG button (logs are saved automatically instead, section 9.1).
- A battery level, and any sound without a buzzer.
- Updating from the phone: the layout has no update slot (update over USB).

The board listens on the **2.4 GHz** band only. A device on 5 GHz Wi-Fi is not heard. The detection lists are not exhaustive.

11.1 Hardware status of this edition

This release (Public Beta 1) has been flashed and run on a real plain ESP32-C6-LCD-1.47 by the project: it starts, shows the interface and finds Wi-Fi networks. The earlier one-button version (3.6-btn) also ran on that board, including the display settings, the LED and reading BOOT as the button. Not every feature has been tried one by one on this release yet (Bluetooth, Zigbee/Thread, the phone page and the saved logs in particular). The release was also built with the real ESP32 toolchain and its interface checked in a desktop simulation (button timing, every screen, the LED and the banners). Please report what you see, good or bad, with the Bug report form.

12. Troubleshooting

The board prints start-up messages over USB: open a serial terminal at 115200 baud on its port. They help a lot when you report a problem.

Problem	Try
The computer does not see the board, or flashing fails	Use a USB **data** cable and another port. Hold BOOT while plugging the cable in, release it, then try again. Add **--baud 115200** if the connection is unreliable.
The screen stays black	Press the button once: the screen may have gone to sleep (LONG on Home turns it off). Unplug, wait 10 seconds, plug in again. Write the .bin again (section 3.1). Check the Serial Monitor for the **free heap** line.
The colours look like a photo negative	Set **LCD_IPS** to 0 in the sketch and upload again.
The picture is shifted sideways or has a coloured edge	Change **LCD_COL_OFS** (34 is the usual value for this 172 px panel; try 0 or 35).
The LED never lights, or the build stops at the LED lines	The sketch uses the core's **rgbLedWrite()**. Update the esp32 board package to 3.3.5 or newer. Check More > Alerts is not LED OFF or DISCREET.

Problem	Try
Nothing happens when I press the button	BOOT is the only button the software reads (RESET cannot be used). A button held while powering up does nothing until it has been let go once. If the screen is off the first press only wakes it.
The button feels too quick or too slow	Change **BTN_SEL_MS** (500) and **BTN_BACK_MS** (1500) in the sketch.
A group says blocked	ZB/TH and the Wi-Fi groups cannot run together (section 7.1). Turn the other side off first.
Wi-Fi finds are missing or slow	Make sure a Wi-Fi group is armed. Wi-Fi pauses while the phone page is on. The list says *Wi-Fi radio: no frames yet* for a few seconds, then it falls back to scanning access points, which sees them in bursts.
Bluetooth finds are slow while a Wi-Fi group is armed	Wi-Fi and Bluetooth share one radio. Arming only Bluetooth groups (BT, TAGS) gives Bluetooth the whole radio.
Log times are blank	The clock is not set: open the phone page's Live view once (section 9.2).
The phone will not join the board's Wi-Fi	Join the exact name shown on the board, which ends in four changing characters, with the password shown. If the phone refuses, forget any older saved network called HUNTR-SEEKR (iPhone: Settings > Wi-Fi > the i next to the name > Forget This Network; Android: touch and hold the name > Forget) and join again. The address is http://192.168.4.1 (not https). Some phones drop networks without internet: turn off "auto-switch to mobile data" while connected.
No sound	The board has no speaker: you need a passive buzzer and **PIN_BUZZER** set to its pin (section 10.2).
A watch or phone shows only "Apple", or does not appear	Normal. Apple devices rotate their address and send very little, so they cannot be told apart.
I want to reset everything	Write the .bin again (section 3.1). That resets the saved settings. Saved logs are kept; wipe them on the phone page.

Still stuck? Open an issue on the project page (Bug report form) with the release name from the boot screen, what you did, and what the Serial Monitor printed.

13. Care, safety and responsible use

- **Power.** Use a proper USB-C supply and cable. If you power the board some other way, follow the maker's safety instructions. Do not cover the board while it runs for long periods.
- **Wiring.** If you add a button, buzzer or GPS module, check the pin numbers against section 10 first. The pins named there as taken are in use by the display, the LED and the BOOT button.
- **The LED is bright.** Do not stare into it from close up; lower **LED_LEVEL** or switch it off in More > Alerts.
- **The hotspot.** The phone page runs a password-protected Wi-Fi hotspot while it is on. Switch it off (HOLD or LONG) when you are done.

Responsible use

This project is meant for defensive, privacy-protecting use: finding out what is around you. Receiving radio signals that are openly broadcast is lawful in many places, but laws differ, and some places restrict particular uses of scanners or of the information you collect. You are responsible for following the laws where you are. **Do not use it to identify, follow or harass individual people.** A match is a clue about a device, not evidence about a person or a place.

Licence

HUNTR/SEEKR is free software released under the **GNU General Public License v3.0**. It comes with **no warranty**. Copyright (C) 2026 SKIrf Labs. It is built with third-party components under their own licences (LGPL, Apache 2.0, BSD, SIL OFL): see THIRD_PARTY_NOTICES.md in the project. Waveshare, Espressif, Arduino and all brand and product names that appear in the detection lists or pictures belong to their owners; this project is not affiliated with, or endorsed by, any of them.

Quick-reference card

TAP (under 0.5 s) **NEXT**: next item.	**HOLD (0.5 to 1.5 s)** **SELECT**: open, start, apply.	**LONG (over 1.5 s)** **BACK**. On Home: screen off.
Start FINDS, SCENE, PAUSE SCAN, MORE. HOLD on **SCENE**, TAP to a scene, HOLD to arm.	**A find** Banner: TAP skips, HOLD tracks. LED pulses in the group colour.	**Track** Closer to zero = closer. TAP = zoom / full. HOLD or LONG leaves.
Phone page More > Phone page. Join the Wi-Fi shown, open **http://192.168.4.1**. Live view sets the clock.	**LED** Teal blip = listening. Colour pulse = find. Faster blink = closer. Violet while held = BACK.	**Blocked** ZB/TH and Wi-Fi groups cannot run together: turn the other side off.
Logs Saved automatically every 5 minutes and when scanning stops. Download them from the phone page.	**Update** Over USB only: write the new .bin at 0x0 with esptool.	**Wrong picture?** **LCD_IPS** 0 or 1 (negative); **LCD_COL_OFS** 0 / 34 / 35 (shifted).

Glossary

Term	Meaning
dBm	Signal strength. Numbers are negative: -40 is very strong, -90 is very weak. Closer to zero means closer.
Scene	A ready-made set of groups armed together (STREET, AIRSPACE, PERSONAL, ZB/TH).
Group	A kind of device the scanner looks for (ALPR, AXON, CAMS, GLASS, DRONE, BT, TAGS, WIFI, ZB/TH).
Arm	Switch a group on so the scanner listens for it.
Banner	The strip that slides in below the title when something new is found.
Hint strip	The three boxes along the bottom that say what TAP, HOLD and LONG do now.
Hotspot	The Wi-Fi network the board makes for your phone while the phone page is on.
Remote ID	The identification and position message many drones broadcast.
Skimmer	A hidden device that copies card details; some use a Bluetooth module with a factory name.
802.15.4	The radio standard under Zigbee and Thread smart-home devices.
Signature	A rule that matches a device by its address, name or codes.